PACKAGE INSERT

LINDE MEDICINAL CARBON DIOXIDE 100% V/V

1. BRAND OR PRODUCT NAME

Linde Medicinal Carbon Dioxide 100%v/v

2. NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)

Medicinal Carbon Dioxide 100%v/v, liquefied medicinal gas, compressed.

There are no excipients.

3. PRODUCT DESCRIPTION

Carbon dioxide is non-flammable, colourless, and odourless gas. It is packaged in gas cylinders as liquified medicinal gas.

Carbon dioxide in compressed gas cylinders is present in both the liquid and gaseous states. Above the liquid in the cylinder is carbon dioxide gas. Only product gas phase is used. e

The physical appearance of container is French Grey colour cylinder shoulder and light French Grey colour cylinder body.

4. PHARMACODYNAMICS

Pharmacotherapeutic group: All other therapeutic agent – medical gases, ATC code: V03AN02

Physico-chemical properties: Carbon dioxide is a colourless, stable and non-flammable odourless gas with a relative molecular weight of 44.01. The relative density of carbon dioxide is 1.53 and therefore it accumulates above the ground. In the gaseous state and at atmospheric pressure, carbon dioxide sublimes in the form of carbon dioxide snow (dry ice) at a temperature of -78.5°C. Carbon dioxide can be liquefied at a temperature of 20°C above 55 bar. Water solubility of carbon dioxide is 833 cm^3/L at normal pressure. At 20°C and 20 bar the solubility increases 14-fold.

Carbon dioxide is the natural component of the air in a volume percentage of 0.035.

Carbon dioxide is the end product of aerobic metabolism and produced in the mitochondria. The partial pressure of carbon dioxide is regulated between narrow limits in the cells of the body and in the organism. Therefore, even a small change of the carbon dioxide partial pressure (pCO_2) can trigger large physiological effect. Carbon dioxide stimulates the respiration by increasing the frequency and volume of respiration. Rapid ventilation occurs by stopping the carbon dioxide administration. The effect on the circulation of carbon dioxide i.e. an increase in heartbeat-volume, heart rate, blood pressure and cardiac-minute-volume, is the result of the impact on the heart and blood vessels, as well as the autonomous nervous system. Carbon dioxide dilates the blood vessels in the brain and it is a strong coronary vasodilator. In therapeutic use of carbon dioxide an already existing central nervous system depression may worsen. A respiratory carbon dioxide concentration above 30-50% will result in carbon dioxide narcosis.

The effects of carbon dioxide accumulation in the body, highly depend on the partial pressure achieved in the blood and tissues, and on the duration and condition of the exposure as well.

Mechanism of action

Inhalation: Carbon dioxide is administered in order to stimulate respiration. The increase in carbon dioxide and the decrease in pH will give raise to a chemoreceptor stimulus and thereby facilitating the spontaneous respiration.

Insufflation: No pharmacological effect is sought. Carbon dioxide is insufflated in order to expand body cavities and provide visibility during investigation and treatment.

Cryosurgery: No pharmacological effect is sought. Carbon dioxide is used only as a freeze media.

Pharmacodynamic effects

Inhalation: Stimulation of respiration.

Insufflation: Distention and thereby creating visibility.

Cryosurgery: Freeze media.

Clinical efficacy and safety

Inhalation: Respiratory stimulation. Risk for carbon dioxide retention and acidosis.

Insufflation: Risks associated with increased volume and increased pressure, and potential for

entrainment to tissues (emphysema or vascular gas emboli).

Cryosurgery: Freeze media.

Paediatric population

The pharmacodynamics properties are the same in all age groups.

5. PHARMACOKINETICS

As carbon dioxide diffuses freely, changes of the partial pressure and the pH value quickly creates intracellular changes in the blood as well. The inhalation of carbon dioxide increases the carbon dioxide partial pressure in arterial blood ($PaCO_2$) and lowers the pH value (respiratory acidosis).

During hyperventilation the $PaCO_2$ falls (hypocapnia), the pH rises, and consequently respiratory alkalosis occur. During the body metabolism in stationary state approx, 200 ml carbon dioxide/min is generated, which is ten times higher than during physical exertion. Carbon dioxide quickly diffuses from the cell into the bloodstream, where it is primarily transported in the form of bicarbonate, or chemically bound to haemoglobin and plasma protein. In solute form (2.4 to 2.7%) the partial pressure is 46 mmHg in the mixed venous blood. When we breathe out, we exhale the carbon dioxide generated in the body. The partial pressure in the alveolus is 40+/5 mmHg in healthy humans, which corresponds to the partial pressure of carbon dioxide in the healthy arterial blood.

All arterial $PaCO_2$ values above 6.1 kPa (46 mmHg) are considered pathological. Healthy persons can exceed this value practically only by inhaling carbon dioxide. When inhaling carbon dioxide, the arterial partial pressure may increase very quickly (30% of carbon dioxide increases the $PaCO_2$ above 27 kPa=200 mmHg). When breathing endogenous carbon dioxide, the increase is limited to a value of approx, 0.4 to 0.8 kPa/min (3-6 mmHg/min). An apnoeic patient has an average of 5 mmHg/min $PaCO_2$.

The carbon dioxide insufflation in laparoscopic procedures will lead to an increase in $PaCO_2$ (approx, 20-40% due to peritoneal resorption, which might be compensated by appropriate lung ventilation.

Absorption

Inhalation: Absorbed in the lungs.

Insufflation: Local distention of organ cavity, minor uptake to blood by diffusion.

Cryosurgery: Not applicable.

Distribution

Inhalation: Carried by the blood and stimulation of chemoreceptors.

Insufflation: Not applicable.

Cryosurgery: Not applicable.

Biotransformation

Inhalation: Exhaled.

Insufflation: Not applicable.

Cryosurgery: Not applicable.

Elimination

Inhalation: Exhaled air.

Insufflation: Gas passage from natural or surgical orphisium.

Cryosurgery: Not applicable.

Linearity/non-linearity

Inhalation: Dose dependent effects.

Insufflation: Not applicable.

Cryosurgery: Not applicable.

Pharmacokinetic/pharmacodynamic relationship(s)

Inhalation: Dose effect related.

Insufflation: Not applicable.

Cryosurgery: Not applicable.

6. INDICATIONS

Carbon dioxide Linde is indicated:

- As additive (5 to 8% CO2) to pure oxygen for stimulation of spontaneous respiration during normobaric administration of oxygen, e.g., for emergency treatment of CO-intoxication or for prevention of hypocapnia in hyperventilation.
- As a gas media in order to create visibility in conjunction with lower gastro-intestinal endoscopic procedures.
- As a gas media in order to create visibility in conjunction with laparoscopic and minimal invasive surgery.
- As insufflation gas during X-ray imaging of hollow organs and cavities, or for angiography.
- As a media for freezing during cryosurgery (e.g. wart removal).

The treatment is indicated in all age group.

7. RECOMMENDED DOSAGE

Posology

When carbon dioxide is used for inhalation, the gas is mixed with oxygen. The concentration of carbon dioxide should be 5-8%. For clinical and experimental studies higher concentrations are possible.

For body cavity examinations carbon dioxide (100%) is used for insufflation.

Carbon dioxide (100%) is used as a freeze media for cryosurgery.

The procedure may be used for minor cutaneous lesions e.g., warts by simple point contact (e.g., cotton tops), or more extensively for treatment of superficial dermal changes (e.g., superficial freeze of larger extension) dedicated cryo-equipment should be used.

For more invasive tissue changes with penetration into deeper layer of tissue (e.g., treatment of cervical cancer and precancerous tissue changes) dedicated cryo-equipment should be used.

Paediatric population

The safety and efficacy of inhalation in children have not been established.

For posology for insufflation and cryosurgery see text above.

8. METHOD OF ADMINISTRATION

Method of administration

Inhalation: The recommended dose is 5% carbon dioxide in oxygen. The amount, the frequency of administration, and the duration of treatment should be adapted individually by an appropriately qualified and experienced specialist. The concentration of the inhaled carbon dioxide should not exceed 8%. The inhalation treatment should only be administered by a specialist. The gas mixture consisting of carbon dioxide and oxygen is produced by mixing the gases with dedicated equipment and should be administered with dedicated equipment used in anaesthesia.

Insufflation: Insufflation should be done by using an automatically controlled insufflation system which, at a minimum, permits continuous display of gas flow and pressure in the insufflated cavity. The amount of gas, the speed, and duration of the insufflation should be individually adapted by the physician responsible for the procedure.

The carbon dioxide insufflation of the thoracic, abdominal, pelvic and lower limb regions is recommended to be performed under general anaesthesia and controlled ventilation. When examining the abdominal cavities a dedicated device should be used. The lowest effective intra-abdominal pressure should be sought, usually not higher than 12-15 mmHg. It is recommended to use the lowest intra-abdominal pressure allowing adequate visualisation of the surgical field.

When insufflating gas into the chest, the internal chest pressure should be set to 6 mmHg and the gas flow to 1.0 l/min. A higher pressure and a higher gas flow can cause mediastinal lesions or acute cardiac output reduction.

The risk of air embolism can be reduced by pre-filling the device at the beginning of the examination. It must be ensured that the carbon dioxide is sufficiently warmed and moistened. Suitable filters on the patient-side outlet of the device should be used for the protection against bacterial infections and gas contaminants. The risk of developing hypercapnia should be acknowledged. Development of hypercapnia may be prevented through appropriate supervision and control measures (i.e., by increasing the respiratory minute volume).

When used for imaging the gas delivery must be done with dedicated device intended for use with carbon dioxide as a contrast media.

Cryosurgery: For simple point contact e.g., cotton tops could be used. For other ways of administration, dedicated equipment for cryosurgery should be used.

Precautions to be taken before handling or administering the medicinal product, see *Section Instruction for Use*.

9. ROUTE OF ADMINISTRATION

Inhalation

10. CONTRAINDICATIONS

There are no absolute contraindications.

11. WARNINGS AND PRECAUTIONS

The carbon dioxide liquefied should only be administered by physician or trained personnel.

The carbon dioxide in the cylinders is under pressure, in liquefied form.

The outlet gas may become liquidised again due to suddenly and fast opening of the valve and contact with the skin may cause frostbite necrosis.

Wearing of proper protective clothing (safety goggles and gloves) is required in the handling and use of liquid medical carbon dioxide. Cylinders containing liquid carbon dioxide must be in vertical position during the use.

Carbon dioxide displaces the oxygen in the air. Proper ventilation must be provided whenever carbon dioxide is used.

A thorough medical assessment should be performed before carbon dioxide is used whenever any of the conditions listed below are present:

- Respiratory disease, limited lung function
- Cardiac arrhythmia
- Coronary artery disease
- Cardiac failure
- Hypovolemia

The blood oxygen saturation should be monitored continuously (e.g., with pulsoximetri) during the intervention.

Inhalation therapy should be avoided in elderly patients with chronic asthma or other lung disease.

During carbon dioxide insufflation of cavities for any stabilization purposes, only the required volume should be administered at which the insufflated volume, speed and duration can be individually identified and controlled.

In hypovolemic patients, the insufflation (leading to capnoperitoneum) must only be done after appropriate volume replacement and with the outmost care because circulatory depression may occur.

Insufflation of the joint cavities must not be performed after bone fractures due to the increased risk of gas embolism.

Paediatric population

There is limited clinical data explicitly addressing the use of carbon dioxide for inhalation, insufflation, and cryosurgery in the paediatric population.

Insufflation in conjunction with laparoscopic surgery is however well-established for paediatric use.

Lower gastro-intestinal endoscopy is clinical routine in children with suspected gastrointestinal disorders such as Crohn's disease. Cryosurgery is also clinical routine in children.

12. INTERACTION WITH OTHER MEDICAMENTS

During concomitant administration of CNS active drugs (e.g. analgesics and anaesthetics) the stimulation of the respiratory centres may fall short due to carbon dioxide intake. This risk exists especially in patients with hypercapnia. The higher blood carbon dioxide concentration level combined with anaesthetics and catecholamines can cause cardiac arrhythmia. Carbon dioxide inhalation can have influence on dosage and effect of muscle relaxants and antihypertensives.

Insufflation of carbon dioxide may also cause a small increase in $PaCO_2$ and subsequent mild stress response, thus possibly facilitating the occurrence of e.g. arrhythmia.

There are no pharmacokinetic drug interactions when carbon dioxide is used in cryosurgery.

Paediatric population

There is no specific data in the paediatric population.

13. PREGNANCY AND LACTATION

Pregnancy

There is very limited experience of use of medical carbon dioxide during pregnancy. Studies in animals have shown no direct or indirect adverse health effects on fertility.

In case there is no medically urgent need, the use of medical carbon dioxide should be avoided during pregnancy as a precautionary measure until further data are available.

As a principal, it is suggested that laparoscopy with the use of capnoperitoneum should be performed only in the second trimester due to the risk of foetal damage. If laparoscopy is performed during the third trimester, monitoring of the foetal heart is strongly recommended.

It is recommended to read the European Association for Endoscopic Surgery (EAES) guidelines.

Breastfeeding

Breast feeding should not take place during or in close connection with carbon dioxide use.

Fertility

There are no studies examining the effects of carbon dioxide use on fertility or early embryonic development.

14. SIDE EFFECTS

Summary of the safety profile

The undesirable effects listed are derived from public domain scientific medical literature and post marketing safety surveillance. Frequencies of the undesirable effects cannot be estimated based on these data.

Inhalation

The most serious side effect is carbon dioxide retention.

The frequencies of the undesirable effects cannot be estimated from the available data.

Insufflation

The most serious side effects are gas/air emboli (CO2), carbon dioxide retention, and intestinal ischaemia.

The frequencies of the undesirable effects cannot be estimated from the available data.

Cryosurgery

There are no expected side effects.

Tabulated summary of adverse reactions

System organ class	Very common (≥1/10)	Common (≥1/100 to <1/10)	Uncommon (≥1/1 000 to 1/100)	Rare (≥1/ 10 000 to 1/1 000)	Very rare (<1/ 10 000)	Not known (cannot be estimated from the available data)
Vascular disorders						*Insufflation:* Gas/air emboli (CO_2)
Respiratory, thoracic and mediastinal disorders						*Inhalation and insufflation:* CO_2 retention, Hypercapnia, Hypoventilation
Injury, poisoning and procedural complications						*Insufflation:* Barotrauma Emphysema Pneumothorax Pneumo-mediastinum
Gastrointestinal disorders						*Insufflation:* Abdominal distension, Intestinal ischaemia

Paediatric population

There are no known additional undesirable effects in the paediatric population than in adults.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorization of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via *NPRA's ADR (Adverse Drug Reactions) reporting website: npra.gov.my [Consumers -> Reporting Side Effects to Medicines (ConSERF) or Vaccines (AEFI)].*

15. SYMPTOMS AND TREATMENT OF OVERDOSE

Inhalation

When inhaling ≤10% carbon dioxide the following symptoms may occur: headache, tinnitus, increased blood pressure, physical irritation, dizziness, and drowsiness. Depending on the concentration an anaesthetic effect associated with loss of consciousness and possibly spasm may also occur.

When inhaling 10-30% carbon dioxide the following symptoms may occur:

- Unconsciousness
- EEG changes, spasm
- Cardiac arrhythmia

Unconsciousness may occur after 1-2 minutes when 20% carbon dioxide is inhaled. Arterial blood pressure with an increase up to 200 mmHg (27 kPa) and cardiac arrhythmias accompanied by EEG changes may occur 25 seconds after inhaling 30%.

Excessive carbon dioxide resorption causes hypercapnia and acidosis during insufflation. The insufficient or missing respiratory compensation can cause acute life-threatening effects on the circulation and gas exchange and can lead to gas embolism in some cases.

Inhalation and Insufflation

In case of sudden occurrence of unusual arrhythmia, systolic and/or diastolic heart murmurs, acute cardio-circulatory depression or a sudden drop in end-expiratory carbon dioxide concentration; gas embolism should be suspected although it occurs rarely. The administration of carbon dioxide should be discontinued immediately, and appropriate medical actions taken (e.g., intubation and controlled ventilation with high alveolar minute volume).

When the venous reverse flow of the lower limb is blocked for a longer time, thrombosis and/or pulmonary embolism are to be expected in rare cases. This risk can be reduced by traditional perioperative thromboembolism prophylaxis and wearing of antithrombotic stockings during the treatment.

After discontinuing administration of carbon dioxide, rapid improvement usually occurs. In typical hypoxic carbon dioxide intoxication, oxygen inhalation together with acidosis correction is required.

There is no specific antidote.

The risk for vascular gas entrainment must be acknowledged, possibly creating gases lock in the heart and subsequent cardio-vascular collapse. In the presence of left-right shunt (e.g., open foramen ovale), stroke can occur.

When used as an intravascular contrast media the potential risk for distal ischaemia must be acknowledged.

Cryosurgery

To liberal use might cause freeze injury.

Paediatric population

The risk for overdose in the paediatric populations is the same as in the adult population.

16. EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

The patient should not show any signs of distress or residual effects before leaving hospital.

17. PRECLINICAL SAFETY DATA

Inhalation

Carbon dioxide is used for stimulation of spontaneous respiration during normobaric administration of oxygen. The arterial carbon dioxide tension represents the balance between the carbon dioxide produced and that eliminated, plus inspired carbon dioxide. Carbon dioxide elimination mainly takes place via ventilation and expired air. If ventilation is increased, the partial pressure of carbon dioxide in the blood will decrease (hypocapnia) and vice versa i.e. if ventilation is decreased, the partial pressure of carbon dioxide in the blood will increase (hypercapnia). By supplementing or reducing carbon dioxide to the inspiratory air in anaesthesia and in intensive care medicine with patients on respirators, arterial or end-tidal-partial pressure of oxygen or partial pressure of carbon dioxide can be kept at desired levels so that blood gas levels do not influence ventilation.

Insufflation

Studies on carbon dioxide pneumo-peritoneum have been conducted in animal species including rats, mice, horses, pigs, dogs and rabbits. Despite the various size and physiological capacity of the experimental models, the effects of increased pressure could be generally described but with variation in intensity depending on animal model used.

Respiratory changes (increased pulmonary artery pressure, acidosis), haemodynamic changes (low cardiac output, increased mean arterial pressure and elevated peripheral resistance), and effects on liver and kidney caused by reduced portal blood flow and increased oxidative stress as measured by circulating biomarkers are reported.

Accidental entering of any gas into the venous system during pneumo-peritoneum supported surgery may cause embolism. The effects from gas emboli seems, from animal data, to support the use of carbon dioxide with a rather high plasma solubility rather than other gases with a lower plasma solubility that seem be associated to a higher risk for fatal outcome.

A review of the potential effects of pneumo-peritoneum independent of insufflated gas, on cancer e.g. by induction of port site metastasis and peritoneal dissemination concluded that there are some advantages and disadvantages with gas induced laparoscopy compared to laparotomy that needs to be taken into consideration in targeted patient populations.

The surgical procedure as such, independent of gas insufflated, is a contributing factor in port site metastasis and release of tumour cells into the systemic circulation in animal studies. Tumour dissemination is worse after carbon dioxide laparoscopy than after laparotomy in animals, whereas precautions to avoid increased tumour dissemination should be taken irrespective which gas used during induction of pneumo-peritoneum. The preclinical literature review supports the clinical recommendation that subjects should be monitored carefully during laparoscopic and other minimal invasive surgery.

Carbon dioxide is used as an alternative to ioided contrast media, is non-allergenic and considered less nephrotoxic than conventional iodinated contrast media for the diagnosis of vascular disease under angiography. There are reported differences in the pathophysiologic response and CNS toxicity to intravascular injections of intravenous injections of carbon dioxide and air in different species, and it is important to consider injection parameters when assessing adverse effects of carbon dioxide in animals. Thus there are no explicit data on the use of carbon dioxide as an alternative contrast media for intra-vascular imaging.

Freeze tissue destruction (Cryo-surgery)

The methodology has been used for decades and no known safety related issues to the use of carbon dioxide as a freezing medium have been found in the preclinical literature.

18. INSTRUCTION FOR USE

General precautions

- Do not smoke or use a naked flame in areas where medicinal gases are administered.
- Medicinal gases must only be used for medicinal purposes.
- Never use grease, oil or similar substances, even if the cylinder valve sticks or if the regulator is difficult to connect.
- Handle valves and devices belonging to them with clean and grease-free hands, i.e. no use of hand-cream etc.
- Never place a mask or nasal prongs directly on textiles while treatment is being carried out since fabrics that become saturated with oxygen can be highly flammable and cause a risk of fire. If this should occur, thoroughly shake and air the textiles.
- In case of cleaning of cylinders or attached equipment, do not use combustible products and especially not oil-based material. In case of doubt, verify compatibility.
- Prior to any use, ensure sufficient quantity of product remains to allow completion of the planned administration.
- Only use standard devices designed for administration of medicinal product.
- When delivered from the manufacturer, the cylinders should have an intact tamper evident seal.

Preparation for use

- Remove the plastic cover from the valve before use.

The instructions below are applicable for cylinders where a separate pressure regulator shall be connected before use:

- Use only regulators designed for product.
- Check that the connection on the coupling or regulator is clean and that the connections are in good condition.
- Never use pliers to force regulators that are designed to be connected manually, as this can damage the connection.
- Check that the regulator is properly attached before opening the valve.
- Open the cylinder valve gently, at least half a turn.
- Check for leakage according to instructions that accompany the regulator.
- In the event of leakage, close the valve and disconnect the regulator. Mark the defective cylinder, keep it separate and return it to your supplier.

Use of cylinders

- Close the cylinder in the event of fire or when not in use.
- Ensure cylinders are secured to a suitable cylinder support in vertical position when in use, to prevent them from falling.
- For cylinders equipped with integrated valves, the user should be prepared to change the cylinder when the pressure gauge is in the yellow zone and change it when it enters the red zone.
- For cylinders that are not equipped with residual pressure valves, the cylinder valve should be closed when a small amount of gas remains in the cylinder (approx. 2 bars). It is important to leave a slight residual pressure in the cylinder in order to protect it from contamination.

- After use, the cylinder valve should be closed with normal force and the regulator or connection depressurised.

Transportation of cylinder

- When being transported, the cylinders must be secured to prevent them from falling.
- Larger cylinders should be transported with appropriate type of trolley. Particular attention should be paid to ensuring connected devices are not accidentally loosened.

Special precautions for disposal and other handling

The cylinder package must not be disposed but returned to the supplier.

19. STORAGE CONDITIONS

- Do not smoke or use a naked flame in areas where medicinal gases are stored.
- Cylinders should be stored in a well-ventilated area reserved for the storage of medicinal gases.
- Cylinders should be stored under cover, kept dry and clean, kept free from oil and grease, free from flammable material.
- Cylinders should be stored at temperatures below 50°C.
- Precautions should be taken to prevent blows or falls.
- Cylinders containing different types of gases should be must be stored separately.
- Full and empty cylinders should be stored separately.
- Cylinders should be stored and transported with valves closed.
- After delivery from the manufacturer a cylinder must have an undamaged plastic valve cover.

20. SHELF LIFE

3 years

21. DOSAGE FORM AND PACKAGING AVAILABLE

Dosage form

Liquified medicinal gas, compressed.

Packaging

The cylinders are made of Aluminium or Steel, equipped Pin-Index or Bullnose valves.

The cylinder body is painted light French Grey and the cylinder shoulder is painted French Grey.

Packs (including material and valves):

Cylinder size (L water capacity)	Valve type	Cylinder material	Content (kg Carbon Dioxide at 15°C,1 bar)
3.1 – 4.5	ISO 407 No.16 / Bullnose BS341-3 No. 8	Aluminium / Steel	2
4.6 – 9.2	ISO 407 No.16 / Bullnose BS341-3 No. 8	Aluminium / Steel	3
9.3 – 13.2	ISO 407 No.16 / Bullnose BS341-3 No. 8	Aluminium / Steel	6
13.3 – 38.5	ISO 407 No.16 / Bullnose BS341-3 No. 8	Aluminium / Steel	9
46.3 – 52.0	ISO 407 No.16 / Bullnose BS341-3 No. 8	Steel	30

Not all pack sizes may be marketed.

22. NAME AND ADDRESS OF MANUFACTURER / PRODUCT REGISTRATION HOLDER

Manufacturer:

Linde Gas Products Malaysia Sdn Bhd (453560-K)
P.O.BOX 10633, GPO KUALA LUMPUR, 50670 WPKL.
No. 1, Jalan Graphite 3, Kawasan Perindustrian Bandar Mahkota Banting,
42700 Banting, Kuala Langat, Selangor Darul Ehsan

Product Registration Holder:

Linde Malaysia Sdn. Bhd (100783-W)
Level 13, The Pinnacle Persiaran Lagoon, Bandar Sunway,
47500 Petaling Jaya, Selangor Darul Ehsan
03-56248888
03-56113512
csc.lg.my@linde.com

23. PRODUCT REGISTRATION NUMBER(S)

MAL24086041X

24. DATE OF REVISION OF THE TEXT

24 Sep 2024

— END —